SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Ravi Teja Y.U (192565088) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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CoI AT-2

Ravi Teja. Y.V

Problem 1: Logistics Company - Backtracking Search

Constraints

B1 Can't be assigned To Route C.

only one Bus per route.

B3 Can't be assigned To Route A.

Every bus must be assigned exactly one route

Step 1: Assign B1 → Route A

Step 2: B_2 → Route B

B_1 → Route A

B_2 → Route B

Step 3: B_3 → Route C

check Constraints:

- B_1 is not in Route C
- B_2 is Before B_3
- one Bus per route.
- B_3 is not Route A
- Every bus assigned once

Final Valid Schedule

Bus	Route
B_1	Route A
B_2	Route B
B3	Route C

Backtracking:

The Algorithm assign buses one by one. After each assignment it checks all Constraints every assignment satisfies, no back tracking is required if failed. The algorithm would undo the last assigment and Try another route.

Problem 2: Campus delivery Robot:

a) • The Robot Can move only in foure direction up, down, left, right.

• The robot Can observe only adjacent cell, So it has incomplete kn to the environment

• Normal movement Cost 1, While pedestrian Zones Cost 3 (1+2 extra).

• Since the environment is partially observable, it updates its map while moving.

b) Solution using uniform cost search

Initial state : Start at S
Goal at G

Step 1: Place start node in priority queue with cost = 0.

Step 2: Expand the node with smallest cost.

Step 3: Ignore blocked construction cell.

Step 4: update the robot knowledge after observing neighbouring cells.

Step 5: Continue expanding the lowest-cost node.

eg: $S \to A = 1$
$A \to B = 1$
$B \to C = 3$ Total = 7
$C \to D = 1$
$D \to G = 1$

c) AI Concept Used:

• The delivery robot acts as an intelligent agent.

• It processes the best action.

• It moves toward the goal.

Environment type:

- Partially obsorwable (Can't see Wholemap).
- Dynamic (New info is discovered while moving).
- Discrete (moves between grid cells).
- Senquential (each action affects future desidison.

D) Justification:

- It always find the minimum - cost path
- It avoids blocked construction Zones.
- It handles different movement costs effectively.
- It Works well in partially obsorwable environment by updating knowledge during navigation.
- It helps the intelligent agent make rational decision and safety reach the goal.